4.19.4.2. BTP GATT Service

The BTP GATT service is composed of a service with three characteristics—C1, C2 and C3 (see Table 34, “BTP GATT service”). The client SHALL exclusively use C1 to initiate BTP sessions by sending BTP handshake requests and send data to the server via GATT ATT_WRITE_REQ PDUs. While a client is subscribed to allow indications over C2, the server SHALL exclusively use C2 to respond to BTP handshake requests and send data to the client via GATT ATT_HANDLE_VALUE_IND PDUs.

Table 34. BTP GATT service

Attribute	Description
BTP Service	UUID = MATTER_BLE_SERVICE_UUID
C1 Characteristic (Client TX Buffer)	UUID = 18EE2EF5-263D-4559-959F-4F9C429F9D11 Characteristic Properties = Write max length = 247 bytes
C2 Characteristic (Client RX Buffer)	UUID = 18EE2EF5-263D-4559-959F-4F9C429F9D12 Characteristic Properties = Indication max length = 247 bytes
C3 Characteristic (Additional commissioning-related data)	UUID = 64630238-8772-45F2-B87D-748A83218F04 Characteristic Properties = Read max length = 512 bytes

For all messages sent from the BTP Client to BTP Server, BTP SHALL use the GATT Write Characteristic Value sub-procedure. For all messages sent from the BTP Server to BTP Client, BTP SHALL use the GATT Indications sub-procedure.

The values of C1 and C2 SHALL both be limited to a maximum length of 247 bytes. This constraint is imposed to align with maximum PDU size when LE Data Packet Length Extensions (DPLE) is enabled on Bluetooth 4.2 hardware. Per Bluetooth® Core Specification 4.2 (amended) Vol 3, Part F, Section 3.2.9 "Long Attribute Values", the maximum characteristic value length is 512 bytes. The maximum lengths of C1 and C2 may increase in a future version of the BTP specification to allow higher throughput on BLE connections whose ATT_MTU exceeds 247 bytes.

C3 is an optional characteristic that the server MAY use to include additional data required during the provisioning as defined in Section 5.4.2.5.7, “GATT-based Additional Data”.

BTP Clients SHALL perform certain GATT operations synchronously, including GATT discovery, subscribe, and unsubscribe operations. GATT discovery, subscribe, or unsubscribe SHALL NOT be initiated while the result of a previous operation remains outstanding. This requirement is imposed by GATT protocol.

4.19.4.3. Session Establishment

Before a BTP session can be initiated, a central SHALL first establish a BLE connection to a peripheral. Once a BLE connection has been formed, centrals SHALL assume the GATT client role for BTP session establishment and data transfer, and peripherals SHALL assume the GATT server role. If peripheral supports LE Data Packet Length Extension (DPLE) feature it SHOULD perform data length update procedure before establishing a GATT connection.

